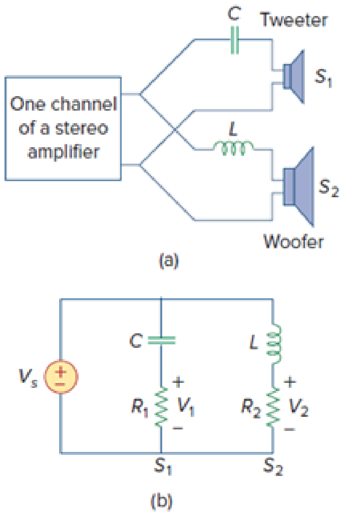
Chapter 14, Problem 19PP

Problem

If each speaker in Fig. 14.66 has an 8-Ω resistance and $C = 10\ \mu$F, find L and the crossover frequency.

Figure 14.66: (a) A crossover network for two loudspeakers, (b) equivalent model



Step-by-step solution

Step 1 of 2

Given that $R_1 = R_2 = 8\Omega$ and $C = 10\mu\text{F}$

For the high pass filter

$$\omega_c = \frac{1}{R_1C}$$

$$2\pi f_c = \frac{1}{R_1C}$$

$$f_c = \frac{1}{2\pi \times R_1 \times C}$$

$$f_c = \frac{1}{2\pi \times 8 \times 10 \times 10^{-6}}$$

$$\therefore \boxed{f_c = 1.989 \text{ k Hz}}$$

Step 2 of 2

For the low pass filter,

$$\omega_c = \frac{R_2}{L}$$

$$2\pi f_c = \frac{R_2}{L}$$

$$L = \frac{R_2}{2\pi f_c}$$

$$L = \frac{8}{2\pi \times 1.989 \times 10^3}$$

$$\therefore \boxed{L = 0.64 \text{ mH}}$$